

LAUNDRY HEAVY DUTY EMULSIFIER

Section: 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Product name : LAUNDRY HEAVY DUTY EMULSIFIER
Product code : 117737E
Use of the Substance/Mixture : Laundry product
Substance type: : Mixture

For professional users only.

Product dilution information : No dilution information provided.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Laundry aid (non-gasing). Automatic process
Laundry aid (non-gasing). Semi automatic process
Recommended restrictions on use : Reserved for industrial and professional use.

1.3 Details of the supplier of the safety data sheet

Company : Ecolab Temizleme Sistemleri Ltd. Şti
Esentepe Mahallesi, Cevizli - Esentepe E-5 Yanyol Caddesi
Vizyon Bulvarı No: 13, Kat 1 No: 65 Turkey TR 34870 KARTAL / İSTANBUL
+90 (216) 458 69 00, Fax: +90 (216) 458 69 07

1.4 Emergency telephone number

Emergency telephone number : +32-(0)3-575-5555 Trans- European
Poison Information Centre telephone number : 114 Ulusal Zehir Danışma Merkezi (UZEM)

Date of Compilation/Revision : 09.04.2020
Version : 1.0

Section: 2. HAZARDS IDENTIFICATION

2.1 Classification of the substance or mixture

Classification (T.R. SEA No 28848)

Acute toxicity, Category 4	H302
Serious eye damage, Category 1	H318
Acute aquatic toxicity, Category 1	H400

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Chronic aquatic toxicity, Category 3

H412

2.2 Label elements

Labelling (T.R. SEA No 28848)

Hazard pictograms :



Signal Word : Danger

Hazard Statements : H302 Harmful if swallowed.
H318 Causes serious eye damage.
H400 Very toxic to aquatic life.
H412 Harmful to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P273 Avoid release to the environment.
P280 Wear protective gloves/ eye protection/ face protection.
Response:
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER/doctor.

Hazardous components which must be listed on the label:
Alcohols, C13-15, branched and linear, ethoxylated
2-[2-(2-butoxyethoxy)ethoxy]ethanol

2.3 Other hazards

None known.

Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

3.2 Mixtures

Hazardous components

Chemical Name	CAS-No. EC-No.	Classification (T.R. SEA No 28848)	Concentration: [%]
Alcohols, C13-15, branched and linear, ethoxylated	157627-86-6	Acute toxicity Category 4; H302 Serious eye damage Category 1; H318 Chronic aquatic toxicity Category 3; H412	>= 30 - < 50
Alcohols, C13-15, branched and linear, ethoxylated	157627-86-6	Acute toxicity Category 4; H302 Eye irritation Category 2; H319 Acute aquatic toxicity Category 1; H400 Chronic aquatic toxicity Category 3; H412	>= 25 - < 30
2-[2-(2-	143-22-6	Serious eye damage Category 1; H318	>= 5 - < 10

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butoxyethoxy)ethoxy]ethanol	205-592-6		
Alcohol ethoxylate	68439-51-0	Chronic aquatic toxicity Category 3; H412	$\geq 2.5 - < 5$
Isopropyl Alcohol	67-63-0 200-661-7	Flammable liquids Category 2; H225 Eye irritation Category 2; H319 Specific target organ toxicity - single exposure Category 3; H336	$\geq 2.5 - < 3$
Cocamidopropyl hydroxysultaine	68139-30-0 268-761-3	Eye irritation Category 2; H319	$\geq 1 - < 2.5$
Amphoteric surfactants	90170-43-7	Eye irritation Category 2; H319	$\geq 1 - < 2.5$
Substances with a workplace exposure limit :			
2-(2-butoxyethoxy)ethanol	112-34-5 203-961-6	Eye irritation Category 2; H319	$\geq 0.1 - < 0.25$

For the full text of the H-Statements mentioned in this Section, see Section 16.

Section: 4. FIRST AID MEASURES

4.1 Description of first aid measures

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- In case of skin contact : Rinse with plenty of water.
- If swallowed : Rinse mouth. Get medical attention if symptoms occur.
- If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

4.3 Indication of immediate medical attention and special treatment needed

- Treatment : Treat symptomatically.

Section: 5. FIREFIGHTING MEASURES

5.1 Extinguishing media

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.

5.2 Special hazards arising from the substance or mixture

- Specific hazards during : Fire Hazard

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firefighting	Keep away from heat and sources of ignition. Flash back possible over considerable distance. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.
Hazardous combustion products	: Depending on combustion properties, decomposition products may include following materials: Carbon oxides nitrogen oxides (NOx)

5.3 Advice for firefighters

Special protective equipment for firefighters	: Use personal protective equipment.
Further information	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Advice for non-emergency personnel	: Ensure adequate ventilation. Remove all sources of ignition. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
Advice for emergency responders	: If specialised clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials.

6.2 Environmental precautions

Environmental precautions	: Do not allow contact with soil, surface or ground water.
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6.3 Methods and materials for containment and cleaning up

Methods for cleaning up	: Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.
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6.4 Reference to other sections

See Section 1 for emergency contact information.
For personal protection see section 8.
See Section 13 for additional waste treatment information.

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Section: 7. HANDLING AND STORAGE

7.1 Precautions for safe handling

- Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Use only with adequate ventilation. Keep away from fire, sparks and heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours). Wash hands thoroughly after handling. Do not breathe spray, vapour. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

7.2 Conditions for safe storage, including any incompatibilities

- Requirements for storage areas and containers : Keep away from heat and sources of ignition. Keep away from oxidizing agents. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Storage temperature : 10 °C to 40 °C

7.3 Specific end uses

- Specific use(s) : Laundry aid (non-gasing). Automatic process
Laundry aid (non-gasing). Semi automatic process

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
2-(2-butoxyethoxy)ethanol	112-34-5	STEL 15 min	15 ppm 101.2 mg/m ³	TR OEL
Further information		EU - I / C (1991 / 322 / EC) list of indicative limit values (updated with 2006 / 15 / EC)		
		TWA (8 Hour)	10 ppm 67.5 mg/m ³	TR OEL
Further information		EU - I / C (1991 / 322 / EC) list of indicative limit values (updated with 2006 / 15 / EC)		

DNEL

Isopropyl Alcohol	:	End Use: Workers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 888 mg/cm ²
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		End Use: Workers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 500 mg/m3 End Use: Consumers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 319 mg/cm2 End Use: Consumers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 89 mg/m3 End Use: Consumers Exposure routes: Ingestion Potential health effects: Long-term systemic effects Value: 26 ppm
2-(2-butoxyethoxy)ethanol	:	End Use: Workers Exposure routes: Inhalation Potential health effects: Short-term - local Value: 101.2 mg/m3 End Use: Workers Exposure routes: Dermal Potential health effects: Long-term systemic effects Value: 20 mg/kg End Use: Workers Exposure routes: Inhalation Potential health effects: Long-term systemic effects Value: 67.5 mg/m3 End Use: Workers Exposure routes: Inhalation Potential health effects: Short-term - local Value: 67.5 mg/m3

PNEC

Isopropyl Alcohol	:	Fresh water Value: 140.9 mg/l Marine water Value: 140.9 mg/l Intermittent use/release Value: 140.9 mg/l Fresh water Value: 552 mg/kg
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		Marine sediment Value: 552 mg/kg Soil Value: 28 mg/kg Sewage treatment plant Value: 2251 mg/l Oral Value: 160 mg/kg
2-(2-butoxyethoxy)ethanol	:	Fresh water Value: 1 mg/l Marine water Value: 0.1 mg/l Intermittent use/release Value: 3.9 mg/l Sewage treatment plant Value: 200 mg/l Sediment Value: 4 mg/kg Soil Value: 0.4 mg/kg Oral Value: 56 mg/kg

8.2 Exposure controls

Appropriate engineering controls

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Individual protection measures

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Eye/face protection (EN 166) : Safety goggles
Face-shield

Hand protection (EN 374) : No special protective equipment required.

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Skin and body protection (EN 14605) : No special protective equipment required.

Respiratory protection (EN 143, 14387) : None required if airborne concentrations are maintained below the exposure limit listed in Exposure Limit Information. Use certified respiratory protection equipment meeting EU requirements(89/656/EEC, (EU) 2016/425), or equivalent, when respiratory risks cannot be avoided or sufficiently limited by technical means of collective protection or by measures, methods or procedures of work organization.

Environmental exposure controls

General advice : Consider the provision of containment around storage vessels.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance	: liquid
Colour	: clear, natural colour
Odour	: alcohol-like
pH	: 7.5 - 9.0, 100 %
Flash point	: 52 °C closed cup, Does not sustain combustion.
Odour Threshold	: Not applicable and/or not determined for the mixture
Melting point/freezing point	: Not applicable and/or not determined for the mixture
Initial boiling point and boiling range	: > 100 °C
Evaporation rate	: Not applicable and/or not determined for the mixture
Flammability (solid, gas)	: Not applicable and/or not determined for the mixture
Upper explosion limit	: Not applicable and/or not determined for the mixture
Lower explosion limit	: Not applicable and/or not determined for the mixture
Vapour pressure	: Not applicable and/or not determined for the mixture
Relative vapour density	: Not applicable and/or not determined for the mixture
Relative density	: 1.0
Water solubility	: soluble
Solubility in other solvents	: Not applicable and/or not determined for the mixture
Partition coefficient: n-octanol/water	: Not applicable and/or not determined for the mixture
Auto-ignition temperature	: Not applicable and/or not determined for the mixture
Thermal decomposition	: Not applicable and/or not determined for the mixture
Viscosity, kinematic	: 82.000 mm ² /s (40 °C)

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Explosive properties : Not applicable and/or not determined for the mixture
Oxidizing properties : The substance or mixture is not classified as oxidizing.

9.2 Other information

Not applicable and/or not determined for the mixture

Section: 10. STABILITY AND REACTIVITY

10.1 Reactivity

No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

No dangerous reaction known under conditions of normal use.

10.4 Conditions to avoid

Heat, flames and sparks.

10.5 Incompatible materials

None known.

10.6 Hazardous decomposition products

Depending on combustion properties, decomposition products may include following materials:
Carbon oxides
nitrogen oxides (NO_x)

Section: 11. TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Product

Acute oral toxicity : Acute toxicity estimate : 905.14 mg/kg
Acute inhalation toxicity : There is no data available for this product.
Acute dermal toxicity : There is no data available for this product.
Skin corrosion/irritation : There is no data available for this product.

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Serious eye damage/eye irritation : There is no data available for this product.

Respiratory or skin sensitization : There is no data available for this product.

Carcinogenicity : There is no data available for this product.

Reproductive effects : There is no data available for this product.

Germ cell mutagenicity : There is no data available for this product.

Teratogenicity : There is no data available for this product.

STOT - single exposure : There is no data available for this product.

STOT - repeated exposure : There is no data available for this product.

Aspiration toxicity : There is no data available for this product.

Components

Acute oral toxicity : Alcohols, C13-15, branched and linear, ethoxylated
LD50 rat: 1,250 mg/kg

2-[2-(2-butoxyethoxy)ethoxy]ethanol
LD50 rat: 6,650 mg/kg

Alcohol ethoxylate
LD50 rat: > 2,000 mg/kg

Isopropyl Alcohol
LD50 rat: 5,840 mg/kg

2-(2-butoxyethoxy)ethanol
LD50 rat: 3,306 mg/kg

Components

Acute inhalation toxicity : 2-[2-(2-butoxyethoxy)ethoxy]ethanol
4 h LC50 rat: > 600 mg/l
Test atmosphere: vapour

Isopropyl Alcohol
4 h LC50 rat: > 30 mg/l
Test atmosphere: vapour

Components

Acute dermal toxicity : Alcohols, C13-15, branched and linear, ethoxylated
LD50 rat: > 2,000 mg/kg

2-[2-(2-butoxyethoxy)ethoxy]ethanol
LD50 rabbit: 3,540 mg/kg

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Alcohol ethoxylate
LD50 rat: > 5,000 mg/kg

Isopropyl Alcohol
LD50 rabbit: 12,870 mg/kg

2-(2-butoxyethoxy)ethanol
LD50 rabbit: 2,764 mg/kg

Potential Health Effects

Eyes : Causes serious eye damage.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Harmful if swallowed.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Eye contact : Redness, Pain, Corrosion

Skin contact : No symptoms known or expected.

Ingestion : No information available.

Inhalation : No symptoms known or expected.

Section: 12. ECOLOGICAL INFORMATION

12.1 Ecotoxicity

Environmental Effects : Very toxic to aquatic life. Harmful to aquatic life with long lasting effects.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : 2-[2-(2-butoxyethoxy)ethoxy]ethanol
96 h LC50 Pimephales promelas (fathead minnow): 2,400 mg/l

Alcohol ethoxylate
48 h LC50 Leuciscus idus (Golden orfe): > 1 mg/l

Isopropyl Alcohol
96 h LC50 Pimephales promelas (fathead minnow): 9,640 mg/l

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Amphoteric surfactants
96 h LC50 Oncorhynchus mykiss (rainbow trout): 4.2 mg/l

2-(2-butoxyethoxy)ethanol
96 h LC50 Fish: 1,300 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Alcohols, C13-15, branched and linear, ethoxylated
48 h EC50 Daphnia magna (Water flea): 0.317 mg/l

2-[2-(2-butoxyethoxy)ethoxy]ethanol
48 h LC50 Daphnia magna (Water flea): > 500 mg/l

Alcohol ethoxylate
24 h EC50 Daphnia magna (Water flea): > 1 mg/l

Isopropyl Alcohol
LC50 Daphnia magna (Water flea): > 10,000 mg/l

Amphoteric surfactants
48 h EC50 Daphnia magna (Water flea): 29 mg/l

Components

Toxicity to algae : 2-[2-(2-butoxyethoxy)ethoxy]ethanol
72 h EC50 Desmodesmus subspicatus (green algae): > 612.6 mg/l

Alcohol ethoxylate
72 h EC50 Desmodesmus subspicatus (green algae): > 1 mg/l

Amphoteric surfactants
72 h EC50 Chlorella vulgaris (Fresh water algae): 9.4 mg/l

12.2 Persistence and degradability

Product

Biodegradability : The surfactants contained in the product are biodegradable according to the requirements of the detergent regulation 648/2004/EC

Components

Biodegradability : Alcohols, C13-15, branched and linear, ethoxylated
Result: Readily biodegradable.

Alcohols, C13-15, branched and linear, ethoxylated
Result: Readily biodegradable.

2-[2-(2-butoxyethoxy)ethoxy]ethanol
Result: Readily biodegradable.

Alcohol ethoxylate
Result: Readily biodegradable.

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Isopropyl Alcohol
Result: Readily biodegradable.

Amphoteric surfactants
Result: Readily biodegradable.

2-(2-butoxyethoxy)ethanol
Result: Readily biodegradable.

12.3 Bioaccumulative potential

no data available

12.4 Mobility in soil

no data available

12.5 Results of PBT and vPvB assessment

Product

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Dispose of in accordance with the European Directives on waste and hazardous waste. Waste codes should be assigned by the user, preferably in discussion with the waste disposal authorities.

13.1 Waste treatment methods

Product	: The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.
Contaminated packaging	: Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.
Guidance for Waste Code selection	: Organic wastes containing dangerous substances. If this product is used in any further processes, the final user must redefine and assign the most appropriate European Waste Catalogue Code. It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste identification and disposal methods in

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compliance with applicable European (EU Directive 2008/98/EC) and local regulations.

Section: 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (ADR/ADN/RID)

- 14.1 UN number : 3082
- 14.2 UN proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Alcohol ethoxylate)
- 14.3 Transport hazard class(es) : 9
- 14.4 Packing group : III
- 14.5 Environmental hazards : Yes
- 14.6 Special precautions for user : None

Air transport (IATA)

- 14.1 UN number : 3082
- 14.2 UN proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(Alcohol ethoxylate)
- 14.3 Transport hazard class(es) : 9
- 14.4 Packing group : III
- 14.5 Environmental hazards : Yes
- 14.6 Special precautions for user : None

Sea transport (IMDG/IMO)

- 14.1 UN number : 3082
- 14.2 UN proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Alcohol ethoxylate)
- 14.3 Transport hazard class(es) : 9
- 14.4 Packing group : III
- 14.5 Environmental hazards : Yes
- 14.6 Special precautions for user : None
- 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code : Not applicable.

Section: 15. REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or

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mixture

according to Detergents : 30 % and more: Non-ionic surfactants
Regulation EC 648/2004 less than 5 %: Amphoteric surfactants, Polycarboxylates

National Regulations

Take note of Dir 94/33/EC on the protection of young people at work.

Other regulations : According to 11 December 2013, Numbered 28848 (Bis), "Ministry of Environment and Forestry"; Regulation on Classification, Labelling and Packaging of Substances and Mixtures.
According to 13 Dec 2014, Numbered 29204, "Ministry of Environment and Urbanization"; Regulation on Safety Data Sheets regarding Dangerous Substances and Mixtures.

15.2 Chemical Safety Assessment

No Chemical Safety Assessment has been carried out on the product.

Section: 16. OTHER INFORMATION

Procedure used to derive the classification according to REGULATION (EC) No 1272/2008 and Regulation T.R. SEA No 28848

Classification	Justification
Acute toxicity 4, H302	Calculation method
Serious eye damage 1, H318	Calculation method
Acute aquatic toxicity 1, H400	Calculation method
Chronic aquatic toxicity 3, H412	Calculation method

Full text of H-Statements

H225 Highly flammable liquid and vapour.
H302 Harmful if swallowed.
H318 Causes serious eye damage.
H319 Causes serious eye irritation.
H336 May cause drowsiness or dizziness.
H400 Very toxic to aquatic life.
H412 Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

ADN – European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR – European Agreement concerning the International Carriage of Dangerous Goods by Road; AICS – Australian Inventory of Chemical Substances; ASTM – American Society for the Testing of Materials; bw – Body weight; CLP – Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR – Carcinogen, Mutagen or Reproductive Toxicant; DIN – Standard of the German Institute for Standardisation; DSL – Domestic Substances List (Canada); ECHA – European Chemicals Agency; EC-Number – European Community number; ECx – Concentration associated with x% response; ELx – Loading rate associated with x% response; EmS – Emergency Schedule; ENCS – Existing and New Chemical Substances (Japan); ErCx – Concentration associated with x% growth rate response; GHS – Globally Harmonized System; GLP – Good Laboratory Practice; IARC – International Agency for Research on Cancer; IATA – International Air Transport Association; IBC – International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in

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Bulk; IC50 – Half maximal inhibitory concentration; ICAO – International Civil Aviation Organization; IECSC – Inventory of Existing Chemical Substances in China; IMDG – International Maritime Dangerous Goods; IMO – International Maritime Organization; ISHL – Industrial Safety and Health Law (Japan); ISO – International Organisation for Standardization; KECI – Korea Existing Chemicals Inventory; LC50 – Lethal Concentration to 50 % of a test population; LD50 – Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL – International Convention for the Prevention of Pollution from Ships; n.o.s. – Not Otherwise Specified; NO(A)EC – No Observed (Adverse) Effect Concentration; NO(A)EL – No Observed (Adverse) Effect Level; NOELR – No Observable Effect Loading Rate; NZIoC – New Zealand Inventory of Chemicals; OECD – Organization for Economic Co-operation and Development; OPPTS – Office of Chemical Safety and Pollution Prevention; PBT – Persistent, Bioaccumulative and Toxic substance; PICCS – Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR – (Quantitative) Structure Activity Relationship; REACH – Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID – Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT – Self-Accelerating Decomposition Temperature; SDS – Safety Data Sheet; TCSI – Taiwan Chemical Substance Inventory; TRGS – Technical Rule for Hazardous Substances; TSCA – Toxic Substances Control Act (United States); UN – United Nations; vPvB – Very Persistent and Very Bioaccumulative

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Numbers quoted in the MSDS are given in the format: 1,000,000 = 1 million and 1,000 = 1 thousand. 0.1 = 1 tenth and 0.001 = 1 thousandth

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.